

Question 26 (a)

Criteria	Marks
• Uses the pedigree chart to explain that the yellow allele is recessive	2
• Provides some relevant information	1

Sample answer:

The inheritance of yellow colour is recessive since both parents are orange but have yellow offspring. The yellow allele must be present in both parents but it is not expressed.

Question 26 (b)

Criteria	Marks
- Explains the possible outcomes of a cross between I and II - Relates differences in the outcomes of the cross to justify the type of inheritance - Communicates information succinctly using appropriate scientific terms and formats	4
- Explains some outcomes of the cross between I and II including sex-linked inheritance - Communicates information logically using appropriate scientific terms and formats	3
Describes an outcome of the cross between I and II through either sex-linked inheritance OR Mendelian genetics	2
Provides any information relevant to the inheritance of colour in the fish	1

Sample answer:

If the inheritance is sex linked, then I and II would be X^AY and X^aX^a respectively. A cross between I and II would result in all male offspring being yellow and all female offspring being orange.

If the inheritance is not sex linked then II would have the genotype aa, whereas I would either have the genotype, AA or Aa. If I was AA then all of the offspring would be orange. But if I was Aa, then 50% of the offspring would be yellow while the other 50% would be orange and the colours would be equally distributed between male and female offspring.

Therefore, the absence of an orange, male fish as a result of the cross between I and II would confirm that the inheritance of colour in the fish is sex linked.

Answers could include:

Punnet squares.